Active Record Optimization Terminology

- Strict Loading — Preventing Lazy Loading, requiring Eager Loading
- Counter Cache — Storing a count in a column and incrementing it at insert time

While in-depth coverage of data normalization is beyond the scope of this chapter, the basics will be covered here including how they were put into practice for Rideshare.

Good data normalization suggests that your design should eliminate duplication of attributes between tables when possible. This means that each database table has a unique set of fields. When data needs to be linked together, Foreign Key columns refer to Primary Key columns and this relationship is both described and enforced using constraints.

Data relationships might take the form of *one to one*, *one to many*, or *many to many*.

Most Rails applications have many data models and those models are related to each other. Active Record code loads data from many tables, which means rows are joined together in various ways. This is where one of the pitfalls can come in, when rows from multiple tables are not joined together efficiently.

In the earlier section you learned about lazy evaluation of Active Record code, which causes Lazy Loading of data. Lazy Loading means that the data is not loaded upfront. Rails has a different term for loading data upfront: *Eager Loading*.

What's the relevance of Eager Loading? The Eager Loading technique is commonly the solution to one of the most common and well known types of inefficient queries, the *N plus one (N+1)* query pattern. As a refresher on the N+1 pattern, it involves queries for at least two tables, where an outer query runs once, and an inner query runs N times.

Let's break it down further. Querying the right side or inner side repeatedly is a problem because usually it's from Lazy Loading data that could otherwise be Eagerly Loaded.

Let's look at an example from Rideshare. Open `bin/rails console` from your Terminal, connecting to the Rideshare database.

Make sure SQL queries are logged to your console and run this code: